DGX Spark Dual Cooler

Screwless 3D-printed cooling cage for two DGX Spark / ASUS Ascent GX10

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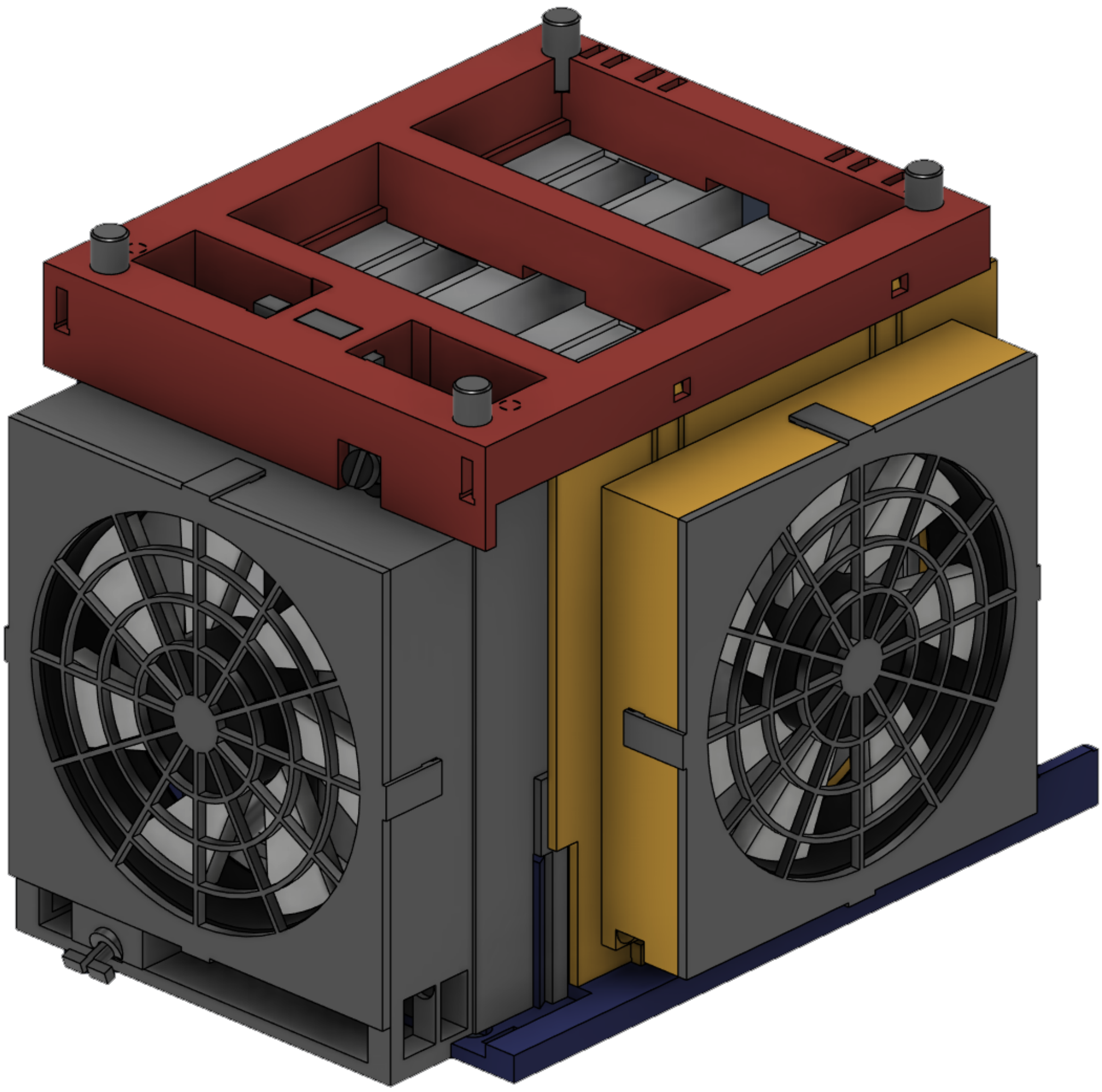


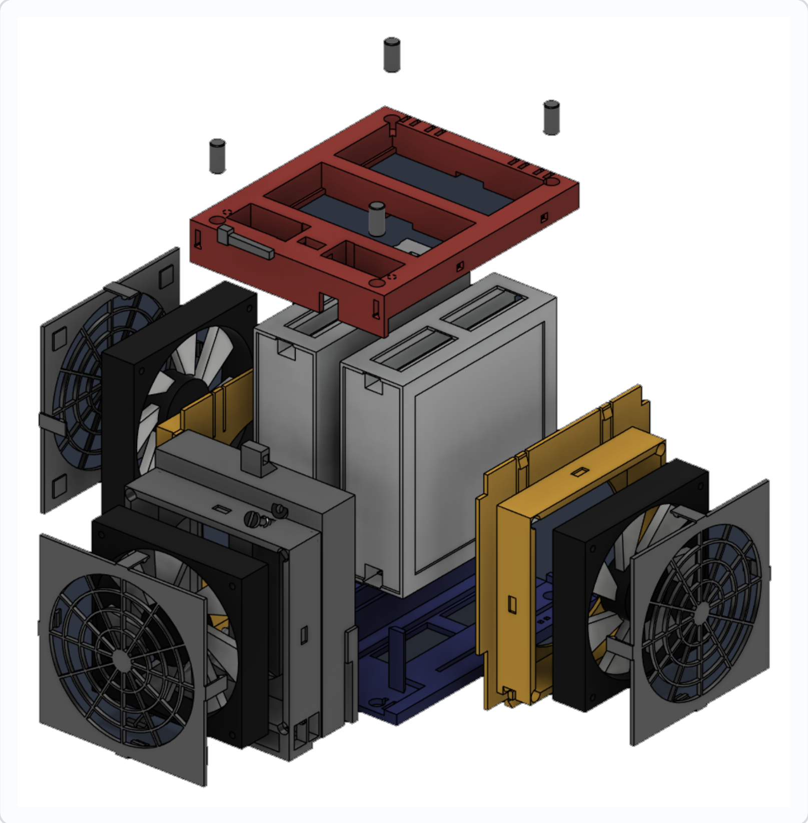
3 x 120 mm	ZERO	206 x 253 x 190	~570 g	15
fans	screws / inserts	mm assembled	PETG @ 20% infill	printed parts

COLOUR KEY — used throughout this guide

- Base tray
- Side fan arms
- Front plenum
- Top brace
- Fans & grilles

Everything clips, slides or wedges together. There is not a single screw, heat-set insert or drop of glue in the assembly. Read pages 2 and 3 before you start printing — the fan you buy decides which grille you print.

What you print



Part	Qty	Size mm	Print orientation	Support	cm3
PRT_01 Base tray	1	176 x 220 x 46	floor down	none	205
PRT_02 Front plenum	1	139 x 66 x 180	back plate down	none *	287
PRT_03 Side fan arm	2	30 x 166 x 172	flat, collar up	none	116 ea
PRT_04 Top brace	1	156 x 190 x 30	top face down	none	238
PRT_05 Fan grille (26 mm fans)	3	135 x 135 x 21	grille face down	none	29 ea
PRT_05B Fan grille (25 mm fans)	3	135 x 135 x 23	grille face down	none	29 ea
PRT_06 Wedge key	1	41 x 6 x 10	on edge	none	1.4
PRT_07 Stacking peg	4	dia 10 x 20	standing	none	1.6 ea
PRT_08 Button pin, long	1	74 long	standing on head	none	1.3
PRT_09 Button pin, short	1	44 long	standing on head	none	0.7

* The plenum reports large overhang area, but every one of those faces is a bridge — the widest unsupported span anywhere in the design is 12.8 mm. No part needs support material. Print orientations above are the measured minimum-overhang orientation for each part.

! **Print only ONE of the two grilles.**
PRT_05 has 1.2 mm clamp pads and suits 26 mm thick fans (e.g. NZXT F120). PRT_05B has 2.2 mm pads and suits 25 mm fans, which includes the Thermalright TL-C12C recommended overleaf. The pads preload the fan against its sealing land; the wrong variant leaves the fan loose and buzzing.

What you buy

Fans — 3 needed

Thermalright TL-C12C-S X3

120 x 120 x 25 mm, 3-pack, 4-pin PWM, S-FDB bearing. 1550 rpm, 66.2 CFM, 1.53 mmH2O, 25.6 dBA, 0.20 A each.

Amazon ASIN B0BN19KSW2

Buy the plain version, not the ARGB one.

Print grille PRT_05B (2.2 mm pads) for these.

Power & speed control — 1 needed

Wathai 36 W fan speed controller

110-240 V AC to 4-12 V DC, 3 A, with a 4-port 3-pin/4-pin splitter. Knob sets voltage; all heat stays in the external brick.

Amazon ASIN B0F4KG6MM1

One brick runs all three fans (0.6 A of 3 A).

A 4th port is spare for a second cage.

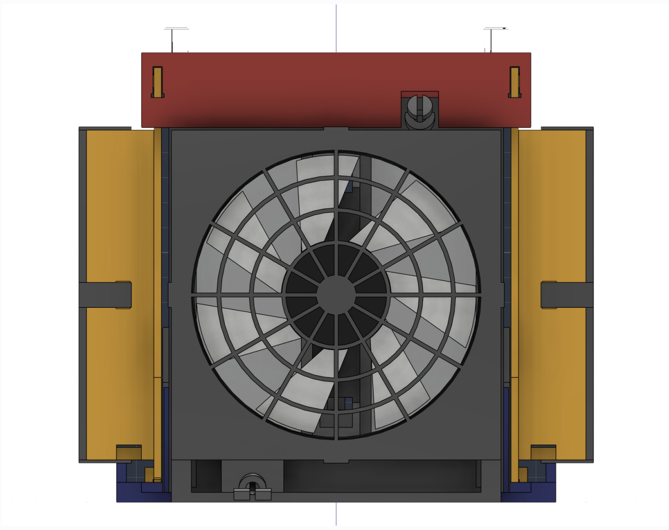
Also needed: about eight 2.5 x 100 mm zip ties for the cable channels. Nothing else — no screws, no inserts, no glue, no wire fan guards.

Print settings

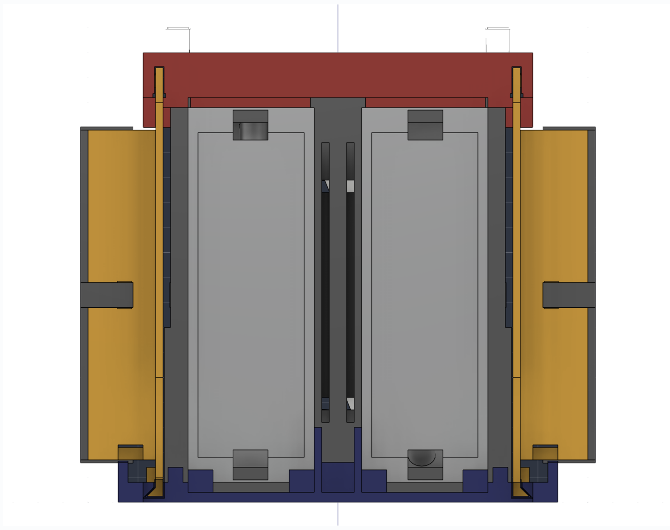
Setting	Value
Material	PETG. All snap-fit strain figures assume E = 2000 MPa.
Nozzle / layer	0.4 mm nozzle, 0.2 mm layers.
Walls	4 perimeters minimum. Every latch, hook and pin leg is a printed spring — thin walls halve their strength.
Infill	20% gyroid is plenty. Nothing here is structural beyond the flexures.
Supports	OFF for every part. Enable bridge/overhang settings instead.
Brim	On for PRT_08, the 74 mm button pin. It stands on a 15 x 4 mm foot.
Bed	248 x 250 mm usable (Bambu P1S). The largest part is the tray at 176 x 220 mm.
Total	About 570 g of PETG for one complete cage.

i **Orient the parts as listed on page 2.**
The orientations in the parts table are not arbitrary — they are the measured minimum-overhang orientation for each part. The top brace in particular prints top-face-down: printed the other way up it needs roughly nine times as much support.

How it holds together



FRONT — intake, front fan, both power-button pins



REAR — ports, QSFP cable, exhaust, cable glands

- The two units are MIRRORED. One is flipped end-for-end so that both filtered bottom panels face outward into the side fans. A consequence is that one unit's power button sits near the top of the front face and the other near the bottom. That is why there are two different button pins.
- Air path: each side fan blows inward through the unit's filtered bottom panel; the front fan pressurises a sealed plenum that feeds both front intakes, plus a centre duct that jets air down the 19 mm gap between the units. Everything exhausts out the rear.
- Nothing is fastened. The arms slide into 45 degree dovetails, the plenum drops onto dovetail posts, the brace latches to the arms, the grilles clip over the fans, and one wedge key locks the brace to the plenum. Take it apart in the reverse order.
- The rear stays clear on purpose. Ports, the QSFP stacking cable and the future exhaust attachment all live there, so allow 95 mm behind the cage as an absolute minimum and 120 mm to be comfortable.

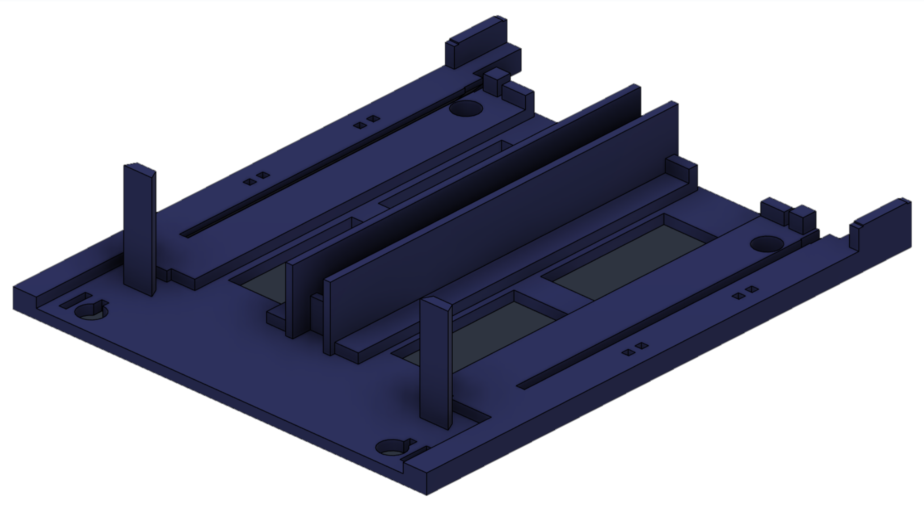
> **! Dry-fit the arms before you commit.**
> The dovetail is a sliding fit with about 0.3 mm of clearance. If your printer runs tight, the arm will stop short. Ease the tongue with a file rather than forcing it — the tray is the one part you do not want to crack.

Assembly

1 Set down the base tray

Put the tray on the desk with the front edge — the end carrying the two dovetail posts — toward you. Every other part locates off this one, so get it square and leave clearance behind it.

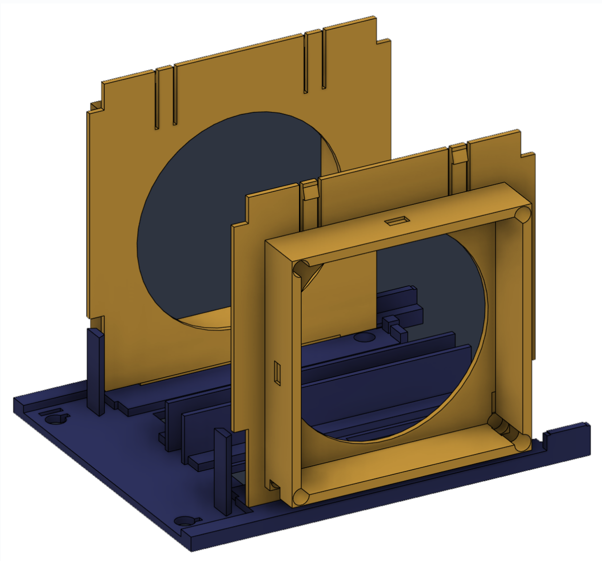
- The rear end is the one with the wide side ears and the two cable glands.
- Nothing is fixed to the desk. The cage simply sits on the tray floor.



2 Slide in the two side fan arms

Both arms are the same printed part, used the second time rotated 180 degrees. Start each arm at the REAR of the tray with its dovetail tongue in the slot, then slide it forward about 150 mm until the tongue bottoms against the closed front end of the slot.

- The square fan collar faces outward on both sides.
- They will feel loose until the top brace goes on in step 7 — that is expected.
- If an arm binds part-way, back it out and check for a stringing artefact in the slot.

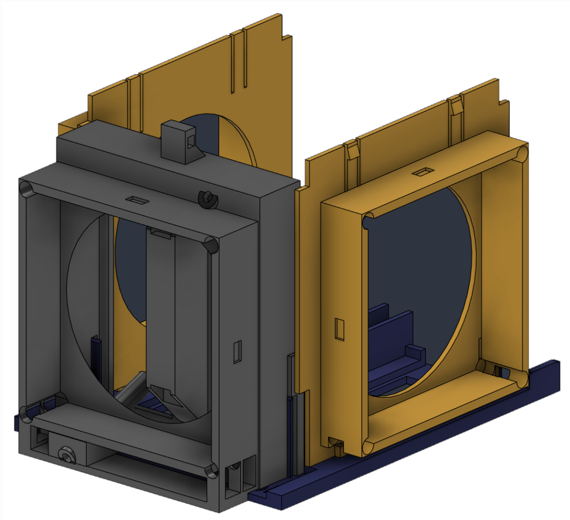


Assembly

3 Drop the front plenum on

Lower the plenum straight down over the two dovetail posts at the front of the tray. It engages only one way round. The last 40 mm should drop under its own weight.

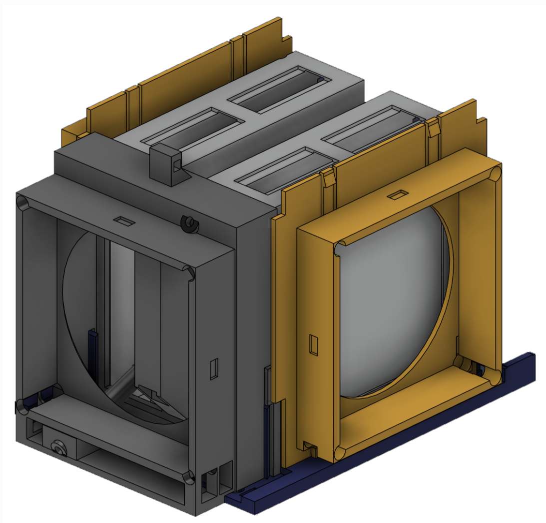
- The dovetails lock it fore-aft and side-to-side; the brace will stop it lifting.
- The two round tube ends visible inside are the power-button pin guides.



4 Lower in the two units

Stand each unit on edge and lower it into its cradle. The second unit goes in flipped end-for-end so that both filtered bottom panels face outward toward the side fans.

- Connect the QSFP stacking cable now — it is far easier before the brace is on.
- Check both front faces sit against the plenum with no gap; that seal is what makes the front fan work.
- One power button will end up high and the other low. Correct.

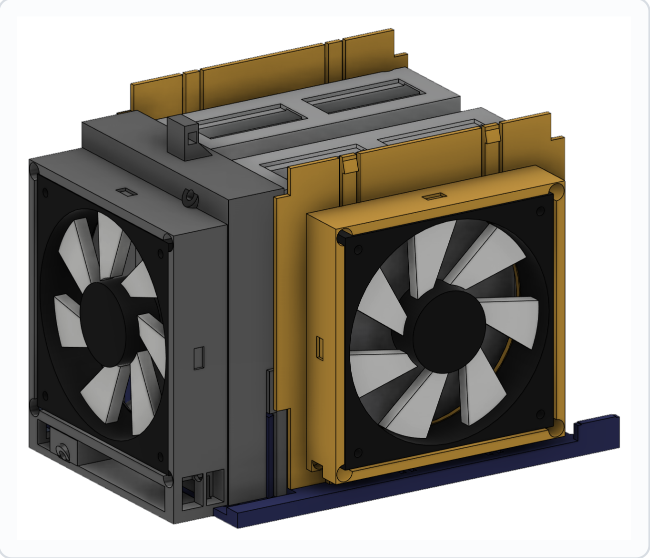


Assembly

5 Push in the three fans

Each fan pushes into its collar from the outside and seats on the round sealing land at the bottom of the pocket. Check the airflow arrow on the fan frame: both side fans must blow INWARD, and the front fan must blow toward the rear.

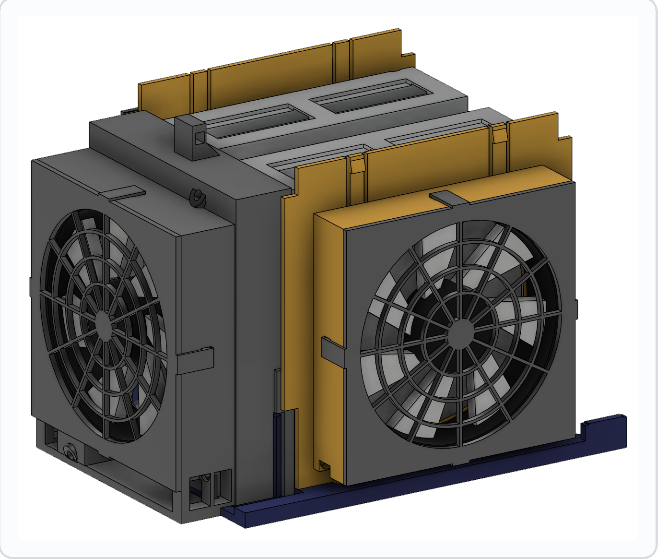
- Drop the lead into whichever corner channel it lands nearest — all four corners are relieved, so any rotation of the fan works.
- The fan should sit about 1 mm proud of the collar face. The grille takes up that gap.



6 Clip on the three grilles

Line each grille up so its four hooks sit over the four notches, then press with both thumbs until all four click home — roughly 11 N, a firm but not violent push.

- Four pads on the inside of the grille preload the fan against its land. If the fan still rattles, you printed the wrong grille variant for your fan thickness.
- To remove later: pry the four hooks outward with a fingernail. The fan comes out without disturbing anything else.

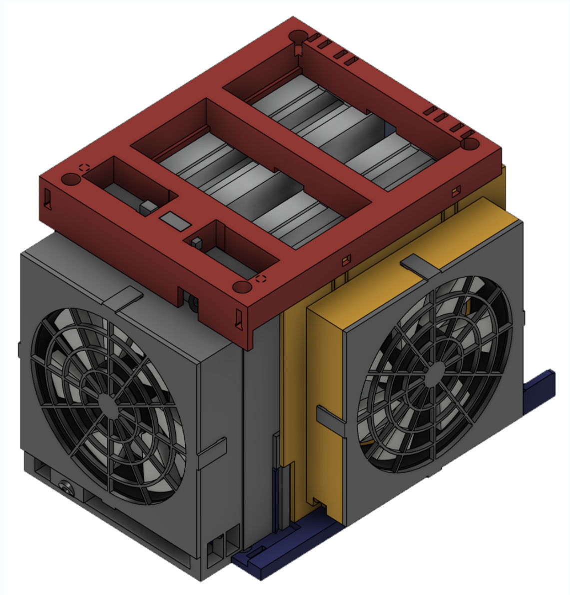


Assembly

7 Fit the top brace, then the wedge key

Lower the brace straight down. The tops of both arms enter the grooves in its underside, and about 5 mm from home the two latch barbs cam over and snap into their catches — expect a distinct click and about 15 N of push. Then slide the wedge key in from the left, through the bridge across the front window and through the plenum tenon, and tap it home.

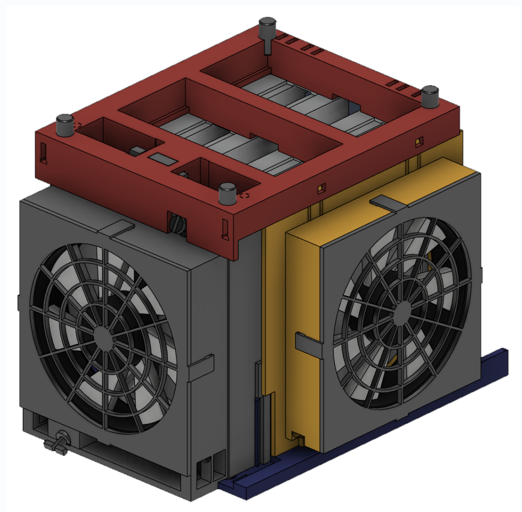
- The key is the only thing resembling a fastener in the whole build. It locks the brace to the plenum.
- It has 0.65 mm of taper, so tap it a little further if the joint ever feels slack.
- To dismantle: drive the key back out, then release both arm latches through the access slots.



8 Stacking pegs and the two button pins

Press the four stacking pegs into the holes in the brace top — only needed if you plan to stack a second cage. Then push each button pin into its tube from outside until it clicks.

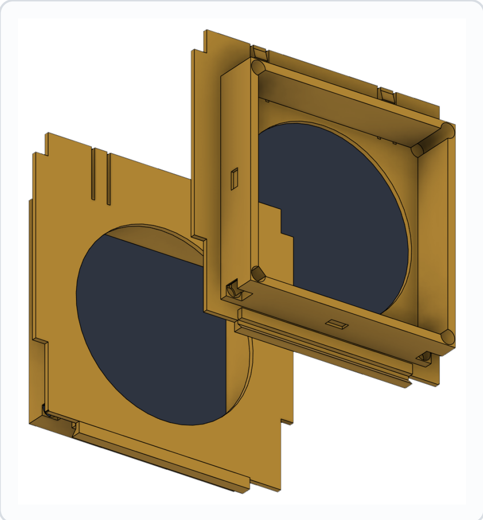
- The long pin (74 mm) goes in the low tube at the bottom front; the short pin (44 mm) in the high tube above the front grille.
- A split barb snaps into a chamber inside the tube. Once in, the pin is captive and slides freely with about 6 mm of stroke.



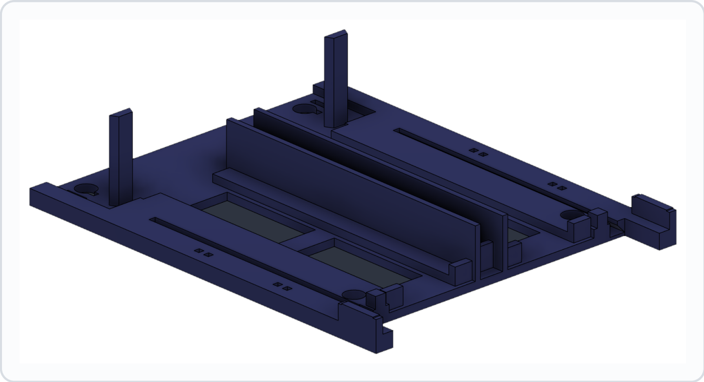
EVERY LEAD IS CHANNELLED, AND EXITS AT THE BACK

Routing the fan leads

All three fan leads run inside the cage and come out together at the rear, where the brick and splitter sit. Nothing crosses the rear ports and nothing sits in the hot exhaust — the run is entirely outboard of the exhaust footprint and below the port strip.

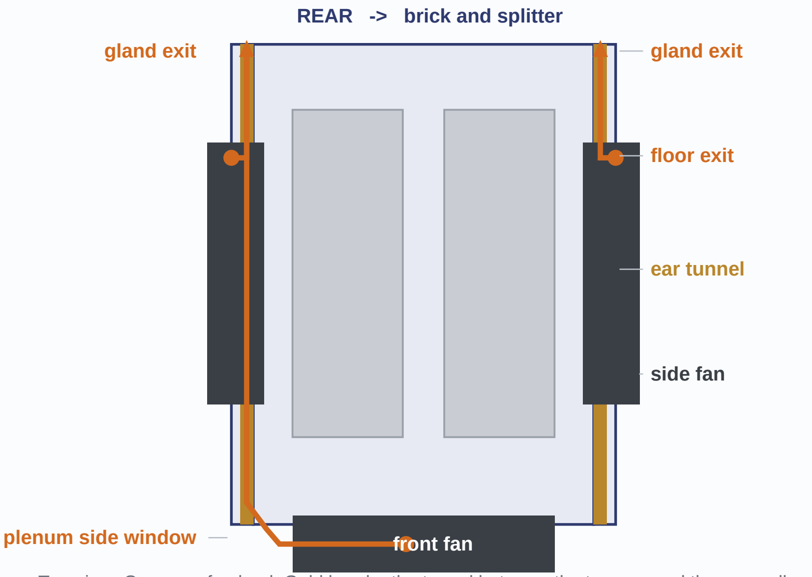


COLLAR — four relieved corners and the floor exit



REAR GLAND — channel out through the tray

- Side fans: the lead drops down whichever pocket corner it sits in, through the floor exit, into the tunnel between the tray ear and the underside of the arm collar.
- Front fan: the lead drops into the plenum skirt and comes out of the side window, level with the front end of the ear, then joins the same tunnel.
- All three then run rearward along the ear tunnel, through the gland in the rear post, and out the back of the tray.
- Zip-tie at the four slot pairs in each ear floor. Leave a little slack at the collar end so a fan can be swapped without cutting ties.



Top view. Orange = fan lead. Gold band = the tunnel between the tray ear and the arm collar.

> **! Leave the rear ears clear above 17 mm.**
> The side arms slide in over the ears during assembly, and the arm collar sweeps that whole band. If you ever add your own bracket back there, keep it under 17 mm tall or you will not be able to get the arms in or out again.

Power, first run and service

Wiring

- Plug all three fans into the Wathai 4-port splitter. The fans are 4-pin PWM but the controller sets speed by voltage — the PWM line simply floats, which is fine.
- Splitter into the 4-12 V brick, brick into the wall. Keep the brick outside the cage; all of its heat stays in the brick, which is the whole reason for choosing it.
- Three fans draw about 0.6 A of the available 3 A, so the fourth port is free for a second cage.

Setting the speed

Start the knob at minimum and wind it up. Below roughly 5 V the fans may not spin up at all — that is normal for voltage control, not a fault. Usable range is about 900 to 1550 rpm. Set it as low as you can while keeping the units happy; the whole point of three 120 mm fans is that they do not have to work hard.

First run checklist

Check	What good looks like
Airflow direction	Both side fans blowing IN, front fan blowing toward the rear. Tissue held at the rear should stream away.
Front seal	Both unit front faces flat against the plenum. A visible gap means the plenum has not dropped fully onto its posts.
Grille clip	All four hooks clicked on each grille. A fan that buzzes is a fan that is not clamped.
Wedge key	Tapped home, brace does not lift at the front corner.
Rear clearance	At least 95 mm behind the cage for the QSFP cable bend. 120 mm is comfortable.
Buttons	Both pins click and spring back freely, and each one reaches its unit's power button.

Service and teardown

- Swap a fan: pry off that grille only. Nothing else has to come apart.
- Full teardown: drive the wedge key out, release both arm latches, lift the brace, lift out the units, lift off the plenum, slide the arms out to the rear.
- Stacking a second cage: fit the four pegs, then sit the next tray on top. Stack pitch is 186 mm.

> i **The centre duct is unproven.**
> The two ram scoops that feed air down the 19 mm gap between the units are a calculated design, not a measured one. Tape over the two slots on the back face of the plenum to run a clean A/B and see whether they actually earn their place.